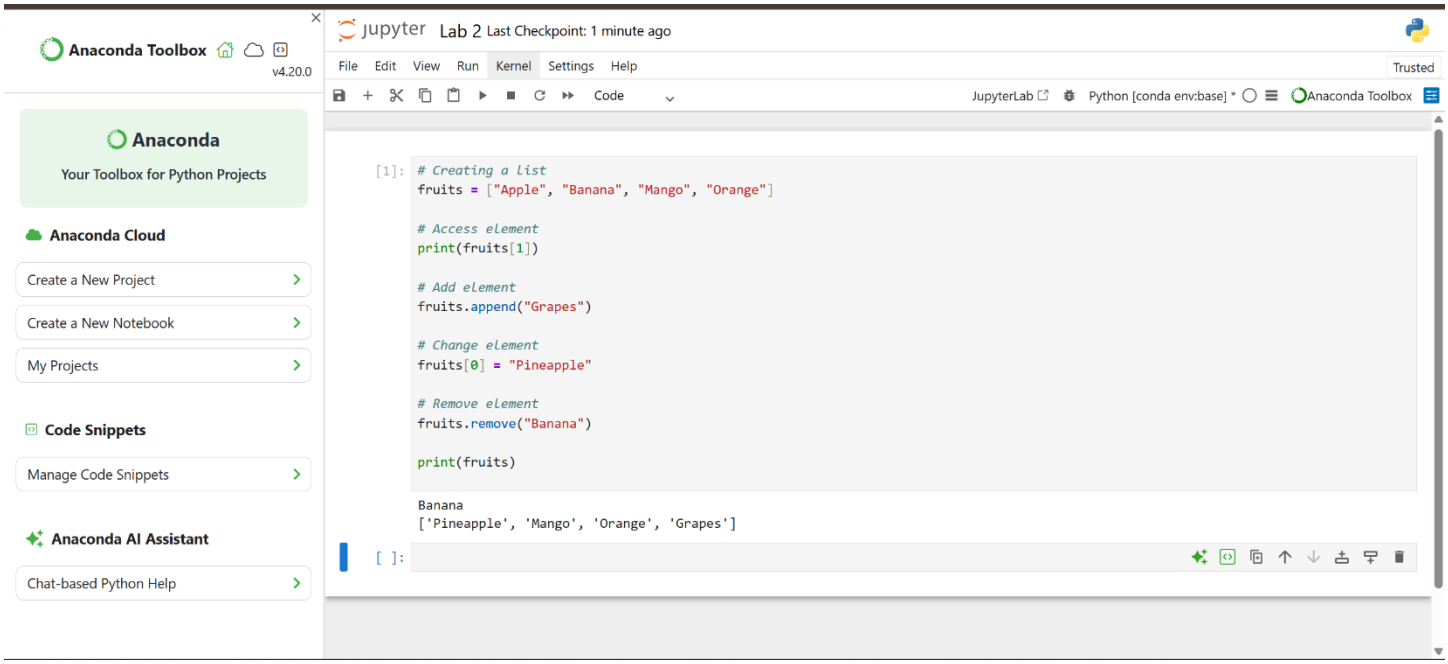
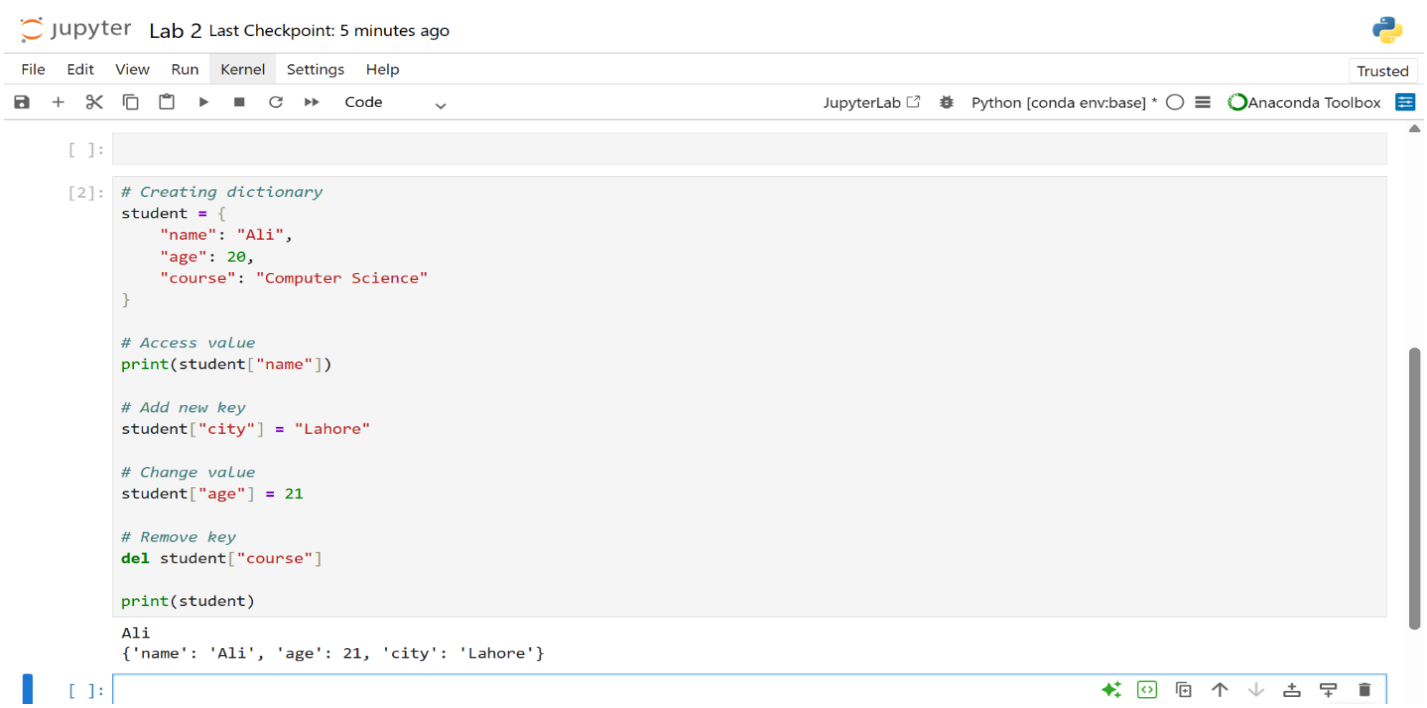
Practical Tasks (M. Athar Ur Rahman)

7. Data Structures:

- List in Python



- Dictionary in Python



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- Tuple in Python:

jupyter Lab 2 Last Checkpoint: 9 minutes ago

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Code JupyterLab Python [conda env:base] * Anaconda Toolbox

```
[3]: # Creating tuple
     numbers = (10, 20, 30, 40)

     # Access element
     print(numbers[2])

     # Length of tuple
     print(len(numbers))
```

```
30
4
```

- Sets in Python

jupyter Lab 2 Last Checkpoint: 10 minutes ago

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Code JupyterLab Python [conda env:base] * Anaconda Toolbox

```
[4]: # Creating set
     colors = {"Red", "Blue", "Green", "Red"}

     # Add element
     colors.add("Yellow")

     # Remove element
     colors.remove("Blue")

     print(colors)
```

```
{'Green', 'Yellow', 'Red'}
```

Practical Tasks (M. Athar Ur Rahman)

- Read CSV file and implement Code:

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[ ]:
```

```
[11]: df= pd.read_csv("PakistanSuicideAttacks Ver 11 (30-November-2017).csv",
                      encoding='latin-1')
```

```
[13]: df
```

[13]:

	S#	Date	Islamic Date	Blast Day Type	Holiday Type	Time	City	Latitude	Longitude	Province	...	Targeted Sect if any	Killed Min	Killed Max	Injured Min	Injured Max	N Sui B
0	1	Sunday-November 19-1995	25 Jumaada al-THaany 1416 A.H	Holiday	Weekend	NaN	Islamabad	33.718000	73.0718	Capital	...	NaN	14.0	15.0	NaN	60	
1	2	Monday-November 6-2000	10 SHa`baan 1421 A.H	Working Day	NaN	NaN	Karachi	24.991800	66.9911	Sindh	...	NaN	NaN	3.0	NaN	3	
2	3	Wednesday-May 8-2002	25 safar 1423 A.H	Working Day	NaN	7:45 AM	Karachi	24.991800	66.9911	Sindh	...	Christian	13.0	15.0	20.0	40	
3	4	Friday-June 14-2002	3 Raby` al-THaany 1423 A.H	Working Day	NaN	11:10:00 AM	Karachi	24.991800	66.9911	Sindh	...	Christian	NaN	12.0	NaN	51	
4	5	Friday-July 4-2003	4 Jumaada al-awal 1424 A.H	Working Day	NaN	NaN	Quetta	30.209500	67.0182	Baluchistan	...	Shiite	44.0	47.0	NaN	65	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
491	492	Thursday-October-05-2017	13 MuHarram 1439 A.H	Working Day	NaN	NaN	Quetta	28.571051	67.496895	Balochistan	...	NaN	21.0	22.0	25.0	NaN	
492	493	Thursday-October-19-2017	27 MuHarram 1439 A.H	Working Day	NaN	8:00 AM	Quetta	NaN	NaN	Balochistan	...	NaN	7.0	7.0	22.0	22	
493	494	Thursday-November-9-2017	19 Safar 1439 A.H	Working Day	NaN	7:00 AM	Quetta	30.221057	67.002524	Balochistan	...	NaN	2.0	2.0	8.0	8	
494	495	Friday-November-24-2017	5 Rabi Al-Awwal 1439 A.H	Working Day	NaN	7:00 AM	Peshawar	33.970623	71.43862	KPK	...	NaN	1.0	3.0	6.0	8	
495	496	Saturday-November-25-2017	6 Rabi Al-Awwal 1439 A.H	Working Day	NaN	7:00 AM	Quetta	30.139626	66.981476	Balochistan	...	NaN	4.0	4.0	16.0	22	

496 rows × 26 columns

```
[14]: df.head()
```

[14]:

	S#	Date	Islamic Date	Blast Day Type	Holiday Type	Time	City	Latitude	Longitude	Province	...	Targeted Sect if any	Killed Min	Killed Max	Injured Min	Injured Max	No. of Suicide Blasts
0	1	Sunday-November 19-1995	25 Jumaada al-THaany 1416 A.H	Holiday	Weekend	NaN	Islamabad	33.7180	73.0718	Capital	...	NaN	14.0	15.0	NaN	60	2.0
1	2	Monday-November 6-2000	10 SHa`baan 1421 A.H	Working Day	NaN	NaN	Karachi	24.9918	66.9911	Sindh	...	NaN	NaN	3.0	NaN	3	1.0
2	3	Wednesday-May 8-2002	25 safar 1423 A.H	Working Day	NaN	7:45 AM	Karachi	24.9918	66.9911	Sindh	...	Christian	13.0	15.0	20.0	40	1.0
3	4	Friday-June 14-2002	3 Raby` al-THaany 1423 A.H	Working Day	NaN	11:10:00 AM	Karachi	24.9918	66.9911	Sindh	...	Christian	NaN	12.0	NaN	51	1.0
4	5	Friday-July 4-2003	4 Jumaada al-awal 1424 A.H	Working Day	NaN	NaN	Quetta	30.2095	67.0182	Baluchistan	...	Shiite	44.0	47.0	NaN	65	1.0

5 rows × 26 columns

```
[15]: df.columns
```

```
[15]: Index(['S#', 'Date', 'Islamic Date', 'Blast Day Type', 'Holiday Type', 'Time',
       'City', 'Latitude', 'Longitude', 'Province', 'Location',
       'Location Category', 'Location Sensitivity', 'Open/Closed Space',
       'Influencing Event/Event', 'Target Type', 'Targeted Sect if any',
       'Killed Min', 'Killed Max', 'Injured Min', 'Injured Max',
       'No. of Suicide Blasts', 'Explosive Weight (max)', 'Hospital Names'
```

Practical Tasks (M. Athar Ur Rahman)

[17]:
```
df.describe()
```

[17]:

	S#	Latitude	Killed Min	Killed Max	Injured Min	No. of Suicide Blasts	Temperature(C)	Temperature(F)
count	496.000000	493.000000	350.000000	480.000000	365.000000	414.000000	491.000000	489.000000
mean	248.500000	32.614705	14.725714	15.206250	31.397260	1.115942	21.111599	69.972579
std	143.327132	2.475917	17.600930	20.270436	38.603842	0.394989	8.369068	15.069622
min	1.000000	24.879503	0.000000	0.000000	0.000000	1.000000	-2.370000	27.734000
25%	124.750000	31.823800	3.000000	3.000000	7.000000	1.000000	14.690000	58.370000
50%	248.500000	33.583300	8.000000	8.000000	20.000000	1.000000	21.405000	70.529000
75%	372.250000	34.004300	20.000000	18.250000	40.000000	1.000000	28.115000	82.499000
max	496.000000	35.383300	125.000000	148.000000	320.000000	4.000000	44.000000	111.000000

[18]:
```
df.shape
```

[18]:
```
(496, 26)
```

[19]:
```
#firstly lets check how many null values are in Blast Day type Column
print("Number of Null Values in 'Blast Day Type' Column: ", df['Blast Day Type'].isnull().sum())

#Null values in the City Column
print("\nNumber of Null Values in 'City' Column: ", df['City'].isnull().sum())

#Null values in the Province Column
print("\nNumber of Null Values in 'Province' Column: ",df['Province'].isnull().sum())

#Number of Null values in the Location Category Column
print("\nNumber of Null Values in 'Location Category' Column: ", df['Location Category'].isnull().sum())

#Number of Null Values in the Location Senstivity Column
print("\nNumber of Null Values in 'Location Senstivity' Column: ", df['Location Sensitivity'].isnull().sum())

#number of null values in target type
print("\nNumber of Null Values in 'Target Type' Column: ", df['Target Type'].isnull().sum())

#number of null values in Target Sect if any
print("\nNumber of Null Values in 'Target Sect if any' Column: ", df['Targeted Sect if any'].isnull().sum())

#number of null values in Killed Min
print("\nNumber of Null Values in 'Killed Min' Column: ", df['Killed Min'].isnull().sum())

#number of null values in Killed Max
print("\nNumber of Null Values in 'Killed Max' Column: ", df['Killed Max'].isnull().sum())

#number of null values in Explosive Weight (max)
print("\nNumber of Null Values in 'Explosive Weight (max)' Column: ", df['Explosive Weight (max)'].isnull().sum())
```

```
Number of Null Values in 'Blast Day Type' Column:  10

Number of Null Values in 'City' Column:  0

Number of Null Values in 'Province' Column:  0

Number of Null Values in 'Location Category' Column:  35

Number of Null Values in 'Location Senstivity' Column:  36

Number of Null Values in 'Target Type' Column:  26

Number of Null Values in 'Target Sect if any' Column:  399

Number of Null Values in 'Killed Min' Column:  146

Number of Null Values in 'Killed Max' Column:  16

Number of Null Values in 'Explosive Weight (max)' Column:  324
```

[ ]: